Mobile-Based Virtual Physics Laboratory with AI Assistance and Interactive Learning for Rural Students

Project ID: R26-IT-150

Project Proposal Report

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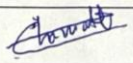
Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Rural school students in Sri Lanka face a persistent gap between theoretical physics knowledge and the ability to explore concepts interactively. The majority of government schools in rural and semi-urban areas lack laboratory infrastructure, curriculum-aligned digital reference materials, and adaptive learning experiences that match the student's specific grade level. Existing virtual laboratory and educational technology platforms either provide static simulations disconnected from the Sri Lankan NIE curriculum, generic search tools not scoped to grade level, or entertainment-focused games that do not produce real physics measurements or enforce conceptual mastery progression.

This research proposes a two-part intelligent learning module within the PhysiV-Lab mobile virtual physics laboratory system, targeting Grade 6-11 students. The first sub-system is an Intelligent Keyword-Based Content Retrieval engine that implements a two-tier architecture: a BM25 baseline for fast keyword matching and a fine-tuned Sentence-BERT semantic search model trained on NIE physics content, both filtered by the student's grade level. The second sub-system is a Curriculum-Aligned 2D Game Generation framework that uses a template-based Unity architecture driven by Firebase configuration data to deliver mastery-based progressive physics gameplay scoped to the NIE syllabus at each grade level, producing real-time physics measurements and formula explanations from student interactions.

Technologies used include Python FastAPI for the backend, rank_bm25 and sentence-transformers for retrieval, Google Colab for model training, Flutter with flutter_unity_widget for mobile integration, Unity 2D for game scenes, and Firebase Firestore for content storage and game configuration. The NIE physics content dataset is constructed manually by the researcher from official NIE textbooks. The SciQ dataset is used for initial domain adaptation of the SBERT model.

Expected outcomes include a deployed grade-adaptive semantic retrieval system validated against a held-out NIE query set, and a functional curriculum-aligned game module with three progressive difficulty stages validated for measurement accuracy, calculation correctness, and grade-difficulty alignment. Together these sub-systems form a Grade-Responsive Syllabus-Aware Intelligent Search and Adaptive 2D Gameplay-Level Scoping Framework unique to Sri Lankan school physics education. The component supports SDG 4 (Quality Education) and SDG 10 (Reduced Inequalities).

Keywords: Semantic Content Retrieval, Sentence-BERT, BM25, Curriculum-Aligned Game Generation, Mastery-Based Progression, Mobile Physics Education, NIE Syllabus

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisor, Mr. Indunil Daluwatte, for his continuous guidance, encouragement, and constructive feedback throughout the development of this research proposal. His insights into educational technology and mobile system design have greatly shaped the direction of this work.

I extend my appreciation to the academic staff of the Department of Information Technology at the Sri Lanka Institute of Information Technology for providing the academic foundation and resources necessary to conduct this research. I am grateful to my fellow team members for their collaboration during system design discussions.

Finally, I wish to express my deepest appreciation to my family and friends for their unwavering support and encouragement throughout my academic journey.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
BM25	Best Match 25 (Information Retrieval Algorithm)
MRR	Mean Reciprocal Rank
NIE	National Institute of Education (Sri Lanka)
NLP	Natural Language Processing
SBERT	Sentence Bidirectional Encoder Representations from Transformers
SDG	Sustainable Development Goal
SUS	System Usability Scale
TF-IDF	Term Frequency – Inverse Document Frequency
UI	User Interface

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1. INTRODUCTION

1.1 Background and Context

Real-World Problem

Physics education in rural Sri Lanka is significantly constrained by a lack of laboratory infrastructure, limited curriculum-aligned digital learning resources, and an absence of adaptive interactive experiences that match the specific grade level of the learner [1], [3]. Students studying physics under the Sri Lankan National Institute of Education (NIE) curriculum for Grades 6 to 11 frequently encounter unfamiliar concepts during self-study with no accessible, curriculum-accurate reference they can query from a mobile device. At the same time, while virtual laboratory platforms and educational games exist globally, none are aligned to the Sri Lankan NIE syllabus, adapted to specific grade levels, or capable of producing real physics measurements from student interactions rather than purely entertainment-driven outcomes.

Two specific gaps define the problem addressed by this component. First, when a student encounters an unfamiliar term such as velocity, pressure, or Ohm's law during a lesson, they have no immediate lightweight search tool scoped to their grade level and curriculum. General internet searches return content at inappropriate difficulty levels and are not aligned to the NIE syllabus. Second, while gamified learning has demonstrated strong potential for improving motivation and retention in science education [1], existing educational games do not convert student gameplay actions into real-time physics measurements, do not enforce mastery-based stage progression, and are not driven by NIE curriculum parameters.

Who is Affected and Why It Matters

The students most significantly affected are the approximately 1.2 million learners enrolled in Grades 6 to 11 in government schools across Sri Lanka, a disproportionate share of whom study in rural schools without laboratory access [3]. Physics consistently registers among the weakest performance areas in national examinations, and the inability to access curriculum-aligned reference materials or engage with concept-reinforcing interactive activities during self-study is a documented contributor to this outcome [3], [5]. Providing lightweight, grade-filtered retrieval and curriculum-aligned gameplay directly addresses both dimensions of this problem within a single mobile application.

Statistical Evidence

Kalogiannakis et al. [1] demonstrate through systematic review of 96 studies that gamification in science education significantly improves both motivation and conceptual retention, particularly where gameplay is directly connected to measurable learning objectives. Elmoazen et al. [5] show that virtual laboratory learning analytics systems consistently improve learning outcomes where student interactions are linked to real physics measurements rather than passive simulation observation. Vijayatheepan [3] documents that virtual laboratory access is unavailable in the majority of rural Sri Lankan government schools, while also noting that teachers express strong demand for curriculum-aligned digital tools that can operate on low-cost mobile devices.

Analysis of Existing Solutions and Their Limitations

PhET Interactive Simulations and Labster provide high-quality virtual experiment environments but are not aligned to the Sri Lankan NIE curriculum, do not offer grade-filtered keyword search, and are not optimised for mobile devices with limited connectivity [5]. General educational search tools such as Khan Academy and Google Search return results across multiple curricula and difficulty levels without grade-level scoping. Educational game platforms such as Kahoot and Quizziz offer gamified quiz formats but do not produce physics measurements from student interactions, do not enforce mastery-based stage progression, and are not driven by curriculum configuration data [1]. Retrieval systems proposed in academic research such as Gil et al. [2] demonstrate keyword-based educational content retrieval but are designed as standalone systems rather than integrated lesson components, and do not support semantic search or NIE curriculum alignment.

SDG Alignment

This component supports SDG 4 (Quality Education) by providing curriculum-aligned retrieval and interactive physics learning for underserved rural students. It supports SDG 10 (Reduced Inequalities) by delivering equal access to intelligent educational tools regardless of geographic location. The research is aligned with the Technology Integration and Management (TIM) research cluster and the Information Technology (IT) specialization.

1.2 Literature Review

A review of twelve recent scholarly sources was conducted to examine the current state of research in educational content retrieval, sentence embedding models, gamification in science education, virtual laboratory learning, and curriculum-aligned game design.

Kalogiannakis et al. [1] provide the most comprehensive systematic review of gamification in science education to date, analysing 96 empirical studies and concluding that gamification significantly improves learner motivation, engagement, and conceptual retention in STEM subjects. Critically, the review identifies that the most effective interventions connect game mechanics directly to measurable scientific outcomes rather than using entertainment as a surface overlay on passive content. This finding directly motivates the measurement-driven game design in this component, where student gameplay actions produce real physics calculations rather than arbitrary scores.

Gil et al. [2] propose a smart retrieval system for digital educational content using machine learning, demonstrating that keyword-based retrieval with ranking algorithms can effectively match student queries to relevant educational materials. Their architecture achieves strong precision at retrieval but does not support semantic understanding of paraphrased queries, does not apply grade-level filtering, and is not designed for integration within a mobile lesson workflow. This limitation motivates the two-tier architecture in this component, where BM25 handles exact matches and fine-tuned SBERT handles semantic paraphrases.

Reimers and Gurevych [4] introduced Sentence-BERT (SBERT), demonstrating that siamese BERT networks produce semantically meaningful sentence embeddings suitable for semantic textual similarity and information retrieval tasks. The all-MiniLM-L6-v2 variant used in this component is an 80MB distilled model validated for fast and accurate semantic search tasks, making it suitable for deployment on a Python backend without GPU requirements. Fine-tuning SBERT on domain-specific

content has been shown to improve retrieval accuracy substantially over the base model in specialised domains [4].

Elmoazen et al. [5] conduct a systematic review of learning analytics in virtual laboratory environments and conclude that the most educationally effective virtual lab systems are those that capture real physics measurements from student interactions and use them to drive formative feedback. Systems that log only activity events without producing measurement-linked outcomes provide insufficient data for learning analytics and do not support meaningful assessment. This finding directly justifies the measurement-driven game design in Sub-System 2, where every student interaction produces a physics calculation that is stored and passed to the analytics module.

Robertson and Zaragoza [10] formally established BM25 as the standard probabilistic ranking algorithm for information retrieval, demonstrating consistent superiority over TF-IDF and providing the theoretical basis for its use as the Tier 1 baseline in this component. The SciQ dataset [8], comprising 13,679 crowd-sourced science questions and answers from school-level subjects, provides the initial domain adaptation corpus for SBERT fine-tuning before NIE-specific fine-tuning is applied. Hugging Face's sentence-transformers library [4] provides the training framework and pre-trained weights.

In the domain of game-based learning, research on mastery-based progression demonstrates that enforcing minimum performance thresholds before unlocking subsequent stages produces significantly stronger conceptual transfer than free-progression designs, particularly in STEM subjects [1], [6]. Template-based game architecture in educational technology has been validated as a scalable approach to curriculum-aligned game delivery, where a small number of reusable game scenes are driven by configuration data rather than requiring new development for each topic or grade level [6]. The Unity game engine combined with the flutter_unity_widget library has been demonstrated as a viable architecture for embedding interactive 2D games within Flutter mobile applications [7].

1.3 Research Gap

Despite advances in educational content retrieval, semantic search models, and game-based science learning, a clear gap exists in the integration of these capabilities within a single curriculum-aligned mobile lesson component for Sri Lankan school physics.

No existing system provides grade-filtered semantic content retrieval aligned to the Sri Lankan NIE physics syllabus. Existing retrieval systems either operate at a general difficulty level across multiple curricula, or use keyword matching without semantic understanding. The combination of a BM25 baseline and a NIE-fine-tuned SBERT semantic model with grade-level filtering has not been proposed or evaluated for school-level physics retrieval in a Sri Lankan or comparable developing-country context.

In the domain of game-based physics learning, no existing platform delivers curriculum-aligned 2D physics games driven by Firebase configuration data that produce real-time physics measurements from student interactions, enforce mastery-based stage progression across three difficulty levels, and scale difficulty parameters automatically to the NIE grade level without requiring Unity code changes. The combination of template-based game architecture, Firebase-driven configuration, measurement-linked student interactions, and NIE curriculum alignment represents a contribution not present in any reviewed system.

Dimension	**Gap in Existing Literature**	**This Study Aims To Investigate**
Content retrieval scope	Retrieval systems are not scoped to NIE curriculum or grade level	Whether grade-filtered BM25 + SBERT retrieval over NIE content

		produces accurate curriculum-aligned results
Semantic understanding	BM25-only systems fail on paraphrased or indirect queries	Whether SBERT fine-tuned on NIE physics content improves retrieval over BM25 baseline for indirect queries
Game measurement linkage	Educational games produce scores without real physics measurements	Whether template-based games producing real-time physics calculations improve conceptual understanding over score-only games
Mastery progression	Existing game platforms allow free progression without threshold enforcement	Whether mastery-based stage gating improves physics concept retention compared to free-progression designs
Grade difficulty scaling	Game difficulty is static or manually adjusted per grade	Whether Firebase-driven dynamic difficulty scaling to NIE grade parameters is feasible without Unity code modification
Curriculum artefact	No structured digital NIE physics content dataset exists for Grades 6-11	Whether a manually curated NIE query-document dataset enables effective SBERT fine-tuning for curriculum-specific retrieval

Table 1:Research Gap Analysis

2. OBJECTIVE

2.1 Main Objective

To design, implement, and evaluate a two-part intelligent learning support module comprising a grade-adaptive two-tier semantic content retrieval system aligned to the Sri Lankan NIE physics curriculum and a curriculum-aligned 2D game generation framework that dynamically scopes gameplay difficulty, physics measurements, and formula explanations to the student's NIE grade level, within a mobile virtual physics laboratory application for Grade 6-11 students in rural Sri Lanka.

2.2 Specific Objectives

1. To construct a structured NIE-aligned physics content dataset comprising 200 to 400 query-document pairs across Grades 6 to 11, to serve as the retrieval corpus for BM25 indexing and SBERT fine-tuning.
2. To implement and evaluate a Tier 1 BM25 keyword retrieval system over the NIE content dataset with grade-level filtering, and establish its performance as a baseline using Precision@K, Recall@K, and MRR metrics.
3. To fine-tune a Sentence-BERT model on the NIE physics content dataset and evaluate whether semantic retrieval improves over the BM25 baseline for indirect and paraphrased student queries.
4. To design and implement a template-based Unity 2D game architecture driven by Firebase configuration data that delivers curriculum-aligned physics gameplay across a minimum of three game templates covering core NIE physics topics.
5. To implement and evaluate a three-stage mastery-based progression system within each game template, enforcing grade-appropriate pass thresholds and producing real-time physics measurements and step-by-step formula explanations from student interactions.
6. To validate the complete two-sub-system component through functional testing, retrieval quality evaluation, game measurement accuracy testing, and usability evaluation with student participants where feasible.

3. METHODOLOGY

3.1 Research Approach

This study adopts a Design Science Research methodology, which is well established in Information Systems research for designing, implementing, and evaluating software artefacts that address identified practical problems [9]. The study will proceed through four phases: problem identification and requirements analysis, artefact construction and system design, prototype implementation, and experimental evaluation. Evaluation will combine functional testing, retrieval quality benchmarking on a held-out query set, game measurement accuracy testing, and usability evaluation with student participants where feasible.

3.2 Sub-System 1: Intelligent Keyword-Based Content Retrieval

NIE Physics Content Dataset Construction

The primary dataset is constructed manually by the researcher from official Sri Lankan NIE Grade 6-11 physics textbooks, freely available at nie.lk. Each document entry covers one physics concept at one grade level and follows a structured schema containing a concept identifier, grade level, topic title, plain-language explanation, formula where applicable, worked example, and a set of keyword tags. For each document, two to four representative query strings are authored to reflect the natural language a student might use when searching for that concept. The target dataset size is 200 to 400 query-document pairs across all six grades and core NIE physics topics. This artefact constitutes an original research contribution as no equivalent structured digital dataset for Sri Lankan secondary physics currently exists in the public domain.

Field	Description
concept_id	Unique identifier for the physics concept
grade	Integer grade level (6 to 11)
topic	NIE lesson topic title
explanation	Plain-language explanation appropriate for the grade level
formula	Physics formula where applicable, with variable definitions
example	One worked numerical example at the appropriate grade level
keywords	Array of keyword tags for BM25 indexing
query_strings	Array of 2 to 4 representative natural language queries

Table 2:NIE Physics Content Dataset Structure

The SciQ dataset [8], comprising 13,679 school-level science questions and answers, is used as a supporting corpus for initial domain adaptation of the SBERT model before NIE-specific fine-tuning is applied. No private or sensitive data is collected or processed.

Tier 1: BM25 Keyword Retrieval

The baseline retrieval approach uses BM25 via the rank_bm25 Python library. The NIE content documents are indexed at startup of the FastAPI backend. When a student submits a query, the backend first applies a grade-level filter to restrict the candidate document set to entries matching the student's grade, then applies BM25 ranking over the filtered set. The top-ranked document is returned if its BM25 score exceeds a defined confidence threshold. This approach requires no model training, runs entirely on CPU, produces reliable results for exact and near-exact keyword queries, and serves as the performance baseline against which SBERT is compared.

Tier 2: Fine-Tuned Sentence-BERT Semantic Retrieval

The enhanced retrieval approach fine-tunes the all-MiniLM-L6-v2 Sentence-BERT model from HuggingFace using the sentence-transformers library. Fine-tuning proceeds in two stages. In Stage 1, the base model is fine-tuned on the SciQ dataset using a MultipleNegativesRankingLoss objective to adapt it to the school science domain. In Stage 2, the domain-adapted model is further fine-tuned on the NIE query-document pairs to specialise it for Sri Lankan curriculum content. Training is conducted on Google Colab free tier, which provides sufficient compute for a dataset of this size without GPU purchase. The trained model weights are exported and deployed on the Python FastAPI backend. At retrieval time, the student query and all grade-filtered candidate documents are encoded using the model, and cosine similarity ranking is applied to return the most semantically relevant result.

Retrieval Decision Logic

At query time, the system applies Tier 1 BM25 first. If the top BM25 score exceeds a defined threshold, the result is returned directly without invoking SBERT, minimising response latency. If BM25 confidence is below threshold, indicating a likely paraphrased or indirect query, Tier 2 SBERT semantic ranking is applied over the grade-filtered candidate set and the cosine similarity top result is returned. This cascaded approach ensures fast responses for common keyword queries while providing semantic fallback for indirect phrasing.

3.3 Sub-System 2: Curriculum-Aligned 2D Game Generation

Template-Based Architecture

Rather than building a separate Unity game for each topic and grade level, the system uses a template-based architecture comprising a small number of reusable Unity 2D game scenes whose content, difficulty parameters, and physics calculations are driven entirely by Firebase Firestore configuration data loaded at runtime. This design means that adding a new topic or adjusting difficulty for a grade requires only a new Firestore document, with no changes to Unity code. Three initial game templates are planned, covering the most frequently assessed physics topics in the NIE curriculum.

Template Name	Physics Concept	Student Interaction
Linear Motion Lab	Speed, Velocity, Distance, Time	Student sets distance or time values; virtual vehicle moves; system calculates and validates speed

Force and Mass Arena	Newton's Second Law, Force, Acceleration	Student applies force to objects of different masses; system measures resulting acceleration and validates F = ma
Pressure Chamber	Fluid Pressure, Area, Force	Student adjusts force or area values on a piston; system calculates resulting pressure and validates P = F/A

Table 3:Game Template Descriptions

Three-Stage Mastery Progression

Every game template implements three progressive stages. Students must achieve the minimum score threshold for each stage before the next stage is unlocked, enforcing mastery-based progression. This design is motivated by research demonstrating that threshold-gated progression produces stronger conceptual transfer than free-progression designs in STEM game-based learning [1].

Stage	Name	Design	Pass Threshold
Stage 1	**Tutorial (Guided)**	All values provided. Formula visible on screen. Student observes real-time physics calculation. No time limit. Hint button available.	No threshold — observational
Stage 2	**Core (Interactive)**	Student controls one or more variables. Calculates result independently. Formula hidden after Stage 1. Immediate feedback per submission.	60%
Stage 3	**Challenge (Independent)**	Student given a target outcome and must adjust multiple variables to achieve it. No formula shown. Countdown timer active. Multiple correct combinations possible.	70%

Table 4: Stage Design Summary

Firebase-Driven Difficulty Scaling

Six game parameters are adjusted based on the student's grade level. All parameter values are stored as Firestore documents and loaded into Unity at runtime via the Python FastAPI backend and Flutter, requiring no Unity code changes to support a new grade or topic configuration.

Parameter	Lower Grades (6-7)	Upper Grades (10-11)
Numerical value ranges	Small whole numbers only	Decimal values and larger magnitudes
Number of controlled variables	One variable at a time	Two or more variables simultaneously
Formula visibility in Stage 2	Formula partially visible	Formula completely hidden
Timer duration in Stage 3	Extended (90 seconds)	Compressed (45 seconds)
Hint availability	Hints available throughout	Hints limited to Stage 1 only
Step-by-step explanation depth	Single formula application	Multi-step derivation with unit analysis

Table 5:Difficulty Scaling Parameters by Grade

End-to-End Implementation Flow

The complete data flow from student action to score storage follows six steps. Flutter sends the student's grade and lesson topic to the Python FastAPI backend. Python queries Firestore and returns the matching game configuration JSON. Flutter launches the appropriate Unity scene via flutter_unity_widget and passes the configuration via postMessage(). Unity parses the configuration, initialises the Stage 1 scene with the correct parameter values, and runs physics calculation validation on each student submission. On stage completion, Unity sends a result JSON object to Flutter via sendMessageToFlutter(). Flutter posts the score and measurement data to Firebase Firestore, where Wijerathna's learning performance evaluation module reads it for mastery level calculation.

3.4 Evaluation Strategy

Evaluation is conducted across four dimensions. Retrieval quality evaluation applies Precision@K, Recall@K, and MRR metrics to both BM25 and SBERT on a held-out set of NIE query-document pairs not used in training. Game functional testing verifies measurement accuracy, calculation correctness, and stage unlock validity. Grade-difficulty alignment is assessed by the researcher and supervisor using a rubric against the NIE curriculum for each grade-topic configuration. Where student participants can be recruited for a pilot session, usability is assessed using the System Usability Scale questionnaire. The following benchmarks are intended targets subject to experimental outcomes:

Evaluation Dimension	Planned Metric	Intended Target
BM25 retrieval precision	Precision@5 on held-out query set	Precision@5 >= 0.70
SBERT retrieval improvement over BM25	MRR comparison on indirect query subset	SBERT MRR >= BM25 MRR on paraphrased queries
Grade-filter accuracy	Proportion of results correctly grade-scoped	>= 95% correctly filtered
Game measurement accuracy	Correctness of physics values generated	100% of physics calculations validated correct
Stage unlock validity	Correct threshold enforcement	Zero incorrect stage unlocks in testing
Grade-difficulty alignment	Rubric score by supervisor	>= 80% of configs rated aligned to NIE grade

Table 6:Planned Evaluation Metrics and Intended Benchmarks

3.5 Technology Stack

- Python FastAPI — backend server handling retrieval queries and game config delivery
- rank_bm25 — BM25 Tier 1 keyword retrieval library
- sentence-transformers — SBERT fine-tuning and Tier 2 semantic retrieval
- HuggingFace all-MiniLM-L6-v2 — lightweight base model for SBERT (80MB)
- Google Colab free tier — SBERT fine-tuning training environment
- Firebase Firestore — NIE content storage, game configuration documents, score storage
- Unity 2D — game scene development and physics calculation engine
- flutter_unity_widget — Unity scene embedding within Flutter mobile application

- Flutter (Dart) — mobile application framework and API client

3.6 Ethical Considerations

All training data is sourced from publicly available datasets (SciQ) and official NIE textbooks freely accessible at nie.lk. No private or personally identifiable student data is collected or processed during development. Student participation in pilot usability testing will be voluntary, and no performance data beyond anonymised session records will be retained for research purposes. Unity game scenes will be tested in an isolated development environment before deployment.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Project Structure and Component Scope

The overall PhysiV-Lab project consists of four team-level components: (1) Grade-Adaptive RAG-Based AI Chatbot and Structured Learning Navigation (Sachintha Nirmal S.M.S, IT22234834), (2) Intelligent Keyword-Based Content Retrieval and Curriculum-Aligned 2D Game Generation (this component, Arthanayake C.T.H, IT22251800), (3) Adaptive User Management and Learning Performance Evaluation (Wijerathna C.D.W, IT22123268), and (4) Interactive Physics Experimentation and Learning (Vidurangi H.I.D.G, IT22143990). This proposal describes component 2 only. Within this component, two independent sub-systems are implemented sharing a common FastAPI backend and Firestore database.

4.2 Component Position in Lesson Workflow

Within the overall lesson workflow, this component provides two services at different points. The Content Retrieval sub-system is accessible to the student throughout the lesson as a reference tool when they encounter an unfamiliar term, complementing Sachintha's AI chatbot which handles conversational lesson navigation. The 2D Game sub-system activates as a standalone lesson activity after the student completes Vidurangi's interactive physics experiments, providing a measurement-driven reinforcement experience before Wijerathna's MCQ assessment. Game scores and physics measurement data are forwarded to Wijerathna's learning analytics module for mastery level calculation.

4.3 System Architecture Diagram

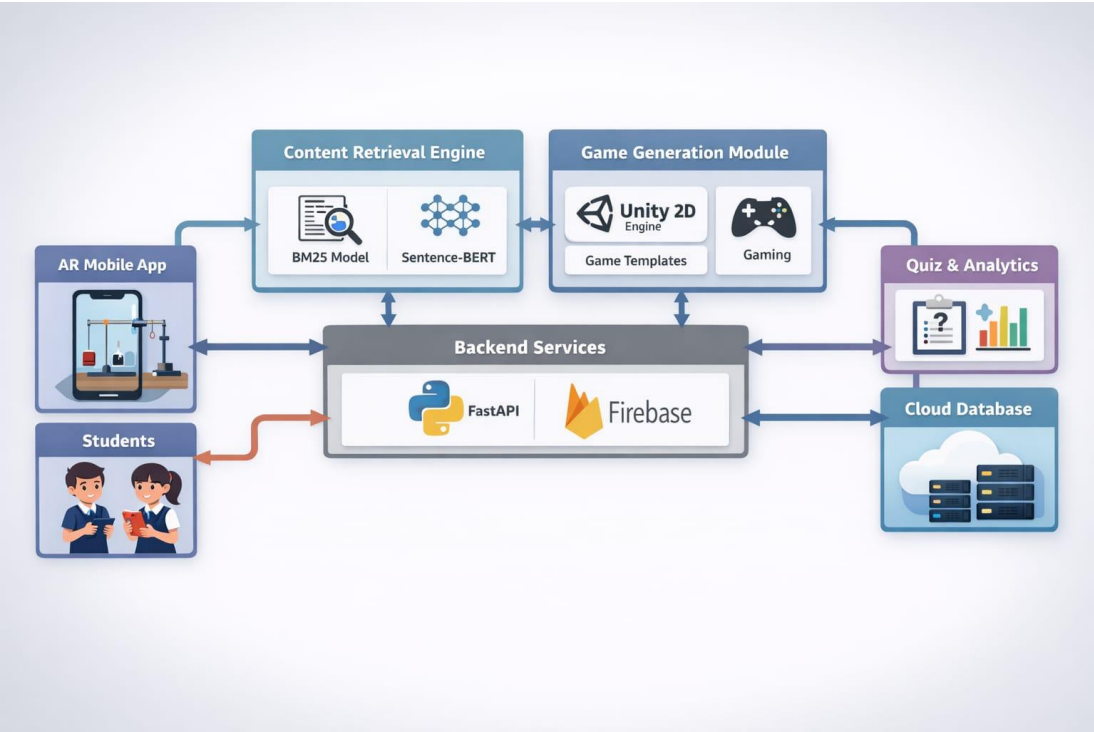


Figure 1:System Architecture Diagram

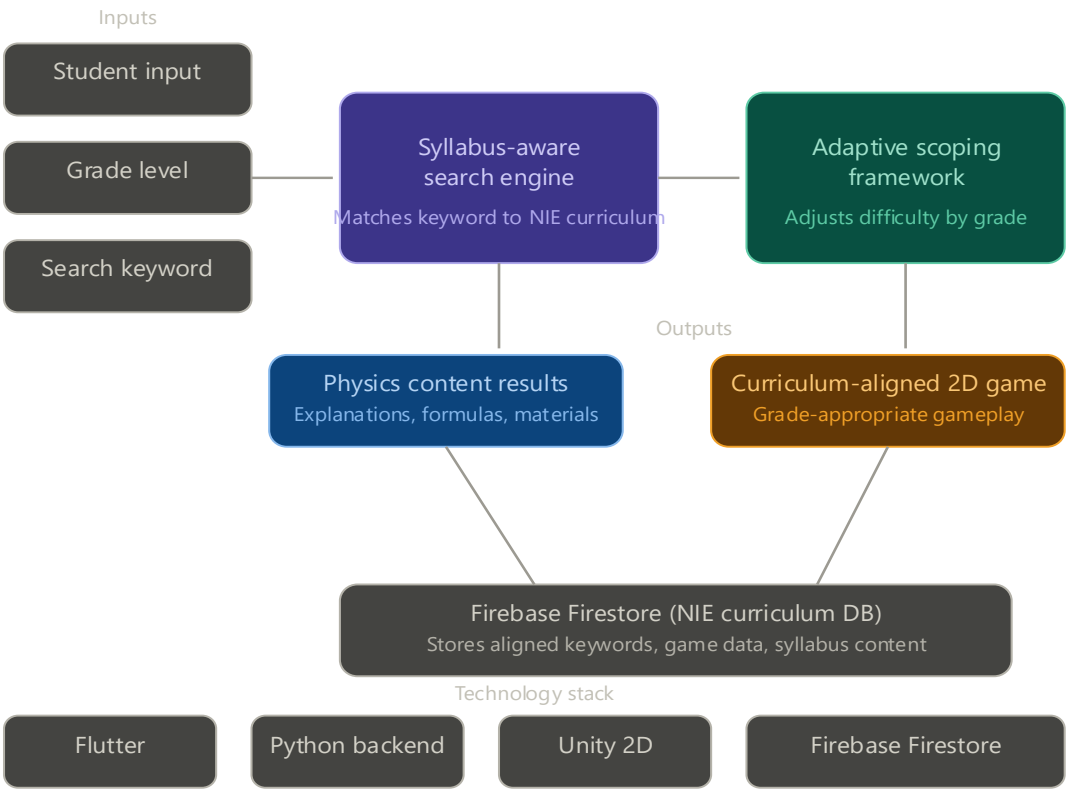


Figure 2: System flow.

4.4 Data Flow

For the retrieval sub-system, the student submits a keyword or natural language query through the Flutter search interface. Flutter sends the query and the student's grade level to the FastAPI POST /retrieve endpoint. The backend applies grade filtering, runs BM25 Tier 1, and invokes SBERT Tier 2 if BM25 confidence is insufficient. The top-ranked result is returned as a JSON object containing the explanation, formula, and example, which Flutter renders on screen.

For the game sub-system, Flutter sends the student's grade and lesson topic to the FastAPI GET /game-config endpoint. The backend queries Firestore and returns the matching game configuration JSON. Flutter launches the Unity scene via flutter_unity_widget, passes the configuration, and receives result JSON from Unity on stage completion. Flutter posts the score and measurement data to Firestore for Wijerathna's module to consume.

4.5 Scalability and Extensibility

The retrieval system is extensible to additional grades, subjects, or curricula by adding new documents to the NIE content Firestore collection. The SBERT model can be retrained on extended datasets without changes to the backend API. The game system is extensible to new topics and grade levels through Firestore configuration documents only, with no Unity code modifications required. The template-based architecture allows new game scenes to be added independently of existing templates.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholder	Role	Primary Interest
Grade 6-11 Students (Rural)	Primary	Accessible keyword search for physics concepts and engaging curriculum-aligned game activities
Physics Teachers	Secondary	NIE curriculum alignment of retrieved content and game difficulty, accurate formula representations
System Integration Team	Primary	Correct API data formatting and score forwarding to analytics module
Research Supervisors	Academic	Validated retrieval metrics, measurable game evaluation outcomes, evidence of novelty
Educational Authorities	Secondary	NIE curriculum accuracy and potential adoption for national digital learning initiatives

Table 7:Stakholder identification

5.2 Functional Requirements

ID	Requirement	Priority
FR-01	The system shall accept keyword or natural language queries from students through the Flutter mobile interface.	Must Have
FR-02	The system shall filter retrieval candidates by the student's grade level before ranking.	Must Have
FR-03	The system shall apply BM25 Tier 1 retrieval and return results when confidence exceeds the defined threshold.	Must Have
FR-04	The system shall apply SBERT Tier 2 semantic retrieval when BM25 confidence is below threshold.	Must Have
FR-05	Retrieved results shall include the physics explanation, formula where applicable, and a worked example.	Must Have
FR-06	The system shall load game configuration from Firestore based on the student's grade and lesson topic.	Must Have
FR-07	The system shall launch the appropriate Unity game template and pass configuration via postMessage().	Must Have
FR-08	Unity game scenes shall implement three progressive stages with mastery-based unlock thresholds.	Must Have
FR-09	Each student game interaction shall produce a real-time physics measurement validated against the correct formula.	Must Have
FR-10	The system shall display step-by-step formula explanations derived from measured values.	Must Have
FR-11	Stage difficulty parameters shall be driven by Firestore configuration without Unity code modification.	Must Have
FR-12	Game scores and measurement data shall be forwarded to Firebase for the analytics module to consume.	Must Have

FR-13	The system shall prevent stage advancement when the student has not met the pass threshold.	Must Have
FR-14	The retrieval system shall return a result within an acceptable response time.	Should Have

Table 8:Functional Requirements

5.3 Non-Functional Requirements

ID	Requirement	Intended Target
NFR-01	The retrieval system shall return a ranked result within an acceptable response time on the backend.	Response within 5 seconds per query
NFR-02	The SBERT model shall run inference on CPU without GPU dependency.	CPU-only inference on FastAPI backend
NFR-03	The game system shall load configuration and launch the Unity scene within an acceptable time.	Scene loaded within 6 seconds on Android
NFR-04	Physics calculation validation in Unity shall produce zero incorrect results.	100% calculation accuracy in functional testing
NFR-05	New game topics and grades shall be addable through Firestore configuration only.	Configuration-only addition verified by test
NFR-06	The system shall handle missing or invalid Firestore game configurations gracefully.	Zero unhandled exceptions in testing
NFR-07	All NIE content retrieved shall be scoped to the student's grade level.	>= 95% grade-filter accuracy

Table 9: Non-Functional Requirements

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market is Grade 6-11 students in rural and semi-urban government schools in Sri Lanka who own Android smartphones but lack access to curriculum-aligned digital reference materials and physics laboratory facilities. Secondary markets include private tutoring centers seeking adaptive physics practice tools, educational technology publishers requiring curriculum-aligned game content for the Sri Lankan market, and government education authorities interested in scalable NIE-aligned digital learning solutions for national examination preparation support.

6.2 Value Proposition

The proposed component provides two distinct value propositions unavailable in any existing platform. For students, it is the only mobile physics tool that provides grade-filtered semantic keyword search aligned to the NIE syllabus, ensuring that retrieved explanations and formulas match the exact curriculum level of the student. For physics learning through games, it is the only platform that converts gameplay actions into real measured physics values, enforces mastery-based stage progression, and scales difficulty automatically to the NIE grade level without requiring curriculum experts to modify game code.

6.3 Revenue Model

Should the PhysiV-Lab platform be commercialized, this component supports a freemium model where core retrieval and basic game templates are available at no cost. Premium features including expanded game template libraries, additional NIE topic configurations, teacher-facing analytics on student game performance, and advanced SBERT retrieval could be offered through institutional subscriptions or government licensing arrangements.

6.4 Competitive Advantage

No existing mobile physics platform for Sri Lankan school students provides NIE-curriculum-aligned semantic search with grade filtering. The template-based Firebase-driven game architecture means new curriculum topics can be added without software development, significantly reducing the marginal cost of content expansion compared to custom-built educational games. The measurement-driven game design produces richer learning analytics data than score-only game platforms, providing a distinct advantage for integration with learning performance evaluation systems.

6.5 Intellectual Property Considerations

The NIE physics content dataset and curriculum-aligned query-document pairs constructed for this research represent original intellectual contributions. The two-tier BM25 plus NIE-fine-tuned SBERT retrieval architecture applied to school-level physics curriculum search may represent a novel contribution if validated. The Firebase-driven grade-scoping game architecture may be protectable as a software design pattern. Appropriate copyright and licensing will be established in consultation with the university's commercialization guidelines following academic publication.

7. BUDGET AND JUSTIFICATION

Category	Item		Qty	Unit (LKR)	Total (LKR)
Software	All planned software (Python, rank_bm25, sentence-transformers, HuggingFace, Unity Personal, Flutter, Firebase SDK) — all open-source or free tier		—	0	0
Cloud Services	Firebase Firestore free tier — sufficient for NIE content dataset and game configs		—	0	0
Compute	SBERT fine-tuning on Google Colab free tier — no GPU purchase needed		—	0	0
Hardware	Development and inference on researcher's personal laptop and Android test device		—	0	0
Contingency	Approximately 10 percent buffer		—	—	0
TOTAL					**LKR 0**

Table 10:Budget summery

Justification

- All software including Python, rank_bm25, sentence-transformers, HuggingFace model weights, Unity Personal edition, Flutter, and Firebase SDK are available under open-source or free-use licenses at zero cost.
- SBERT model fine-tuning is conducted on Google Colab free tier, which provides sufficient GPU compute for a dataset of 200 to 400 training pairs without purchase.
- Firebase Firestore free tier provides sufficient read and write capacity for the NIE content dataset and game configuration documents at the prototype scale.
- The NIE physics content dataset is constructed from publicly available NIE textbooks at nie.lk, incurring no data acquisition cost.
- Unity Personal edition is available at no cost for non-commercial research projects meeting the revenue threshold, which this academic prototype satisfies.

8. WORK BREAKDOWN STRUCTURE (WBS)

WBS ID	Planned Activity	Planned Deliverable	Est. Duration
1.0	**Requirements and Design**		
1.1	Define NIE content schema and target concept list for all six grades	Content schema specification	3 days
1.2	Design retrieval API interface and game config schema	API and schema documents	3 days
1.3	Define game template designs and Firebase config structure	Game design document	4 days
2.0	**Dataset and Artefact Construction**		
2.1	Extract and structure NIE physics content from textbooks (Grades 6-11)	200-400 entry NIE dataset in Firestore	2 weeks
2.2	Author query strings for each NIE dataset document	Complete query-document pairs	1 week
3.0	**Sub-System 1: Retrieval Development**		
3.1	Implement BM25 retrieval with grade filtering and evaluate as baseline	BM25 baseline with evaluation results	1 week
3.2	Fine-tune SBERT on SciQ for domain adaptation (Google Colab)	Domain-adapted SBERT model	1 week
3.3	Fine-tune SBERT on NIE query-document pairs	NIE-specialised SBERT model	1 week
3.4	Implement cascaded Tier 1 / Tier 2 retrieval logic in FastAPI	Working two-tier retrieval API	4 days
3.5	Evaluate retrieval system on held-out NIE query set	Retrieval evaluation report	4 days
4.0	**Sub-System 2: Game Development**		
4.1	Build Unity Linear Motion Lab template with stage logic and calculation validation	Working game template 1	1.5 weeks
4.2	Build Unity Force and Mass Arena template	Working game template 2	1 week
4.3	Build Unity Pressure Chamber template	Working game template 3	1 week

4.4	Implement Firebase config loading and grade-parameter injection in Unity	Firebase-driven config system	4 days
4.5	Implement flutter_unity_widget integration and result forwarding to Firestore	Flutter-Unity integration	4 days
5.0	**Integration and Testing**		
5.1	Integrate both sub-systems with FastAPI backend and Flutter app	Integrated prototype	1 week
5.2	Functional testing of retrieval pipeline and game measurement accuracy	Functional test report	1 week
5.3	Usability pilot with student participants where feasible	Usability evaluation report	1 week
6.0	**Documentation and Reporting**		
6.1	Write research report chapter for this component	Draft report chapter	2 weeks
6.2	Prepare presentation materials	Presentation slides	3 days

Table 11: Work Breakdown Structure

9. GANTT CHART

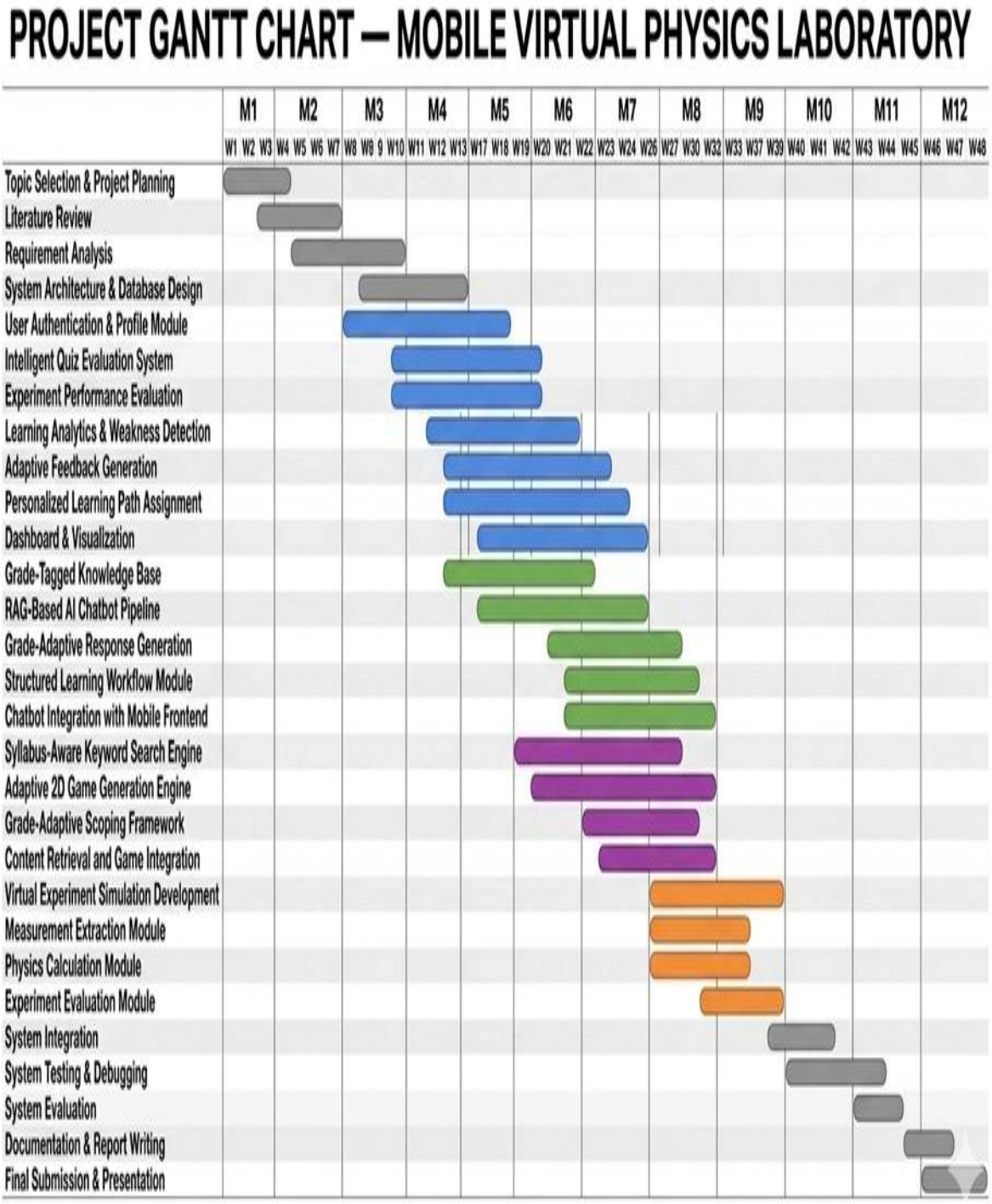


Figure 3: Grant chart

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11. APPENDICES

Appendix A: Preliminary Feasibility Study

To assess technical viability, a preliminary investigation was conducted into the availability of NIE physics content and the behaviour of BM25 retrieval on a small manually authored sample. A set of 30 query-document pairs was constructed from Grade 8 NIE physics topics covering speed, force, and pressure. BM25 was applied to these 30 documents using rank_bm25, and retrieval was tested with 15 direct and 10 indirect query strings. Preliminary observations indicate that BM25 achieves strong precision for direct keyword queries (exact concept terms) but misranks several indirect paraphrased queries, motivating the addition of the SBERT Tier 2 semantic fallback. These are preliminary observations based on a small manually curated sample and should not be interpreted as validated final retrieval metrics.

A preliminary Unity prototype was also constructed for the Linear Motion Lab template to verify that the flutter_unity_widget integration is functional on an Android test device and that the postMessage and sendMessageToFlutter communication channels operate correctly. The prototype confirms that game configuration data can be injected from Flutter into Unity at runtime and that physics calculation results can be forwarded back to Flutter for Firebase storage. This supports the technical feasibility of the template-based Firebase-driven game architecture.

Appendix B: NIE Physics Content Dataset Sample Entries

The following illustrates the planned structure and content for two sample entries from the NIE physics content dataset:

Field	Grade	Content
concept_id	8	speed_grade8
topic	8	Speed, Distance and Time
explanation	8	Speed is how fast an object moves. It is calculated by dividing the distance travelled by the time taken. If a car travels 120 km in 2 hours, its speed is 60 km/h.
formula	8	speed = distance / time (v = s / t) where v is speed in m/s, s is distance in metres, t is time in seconds
example	8	A cyclist covers 500 m in 25 seconds. Speed = 500 / 25 = 20 m/s.
keywords	8	speed, distance, time, velocity, fast, slow, calculation, formula

query_strings	8	['how do you calculate speed', 'what is the formula for speed', 'speed distance time', 'how fast is something moving']

Equation 1:NIE physics content dataset

Appendix C: Firebase Game Configuration Schema Example

The following shows the Grade 8 Speed configuration document stored in Firestore for the Linear Motion Lab game template:

Field	Value (Grade 8 Speed)
template_id	linear_motion_lab
grade	8
topic	Speed
formula	v = s / t
stage1_distance_range	[100, 500] metres (whole numbers)
stage1_time_range	[10, 50] seconds (whole numbers)
stage2_controlled_var	distance
stage2_formula_visible	false
stage2_pass_threshold	0.60
stage3_target_speed	randomly selected from [15, 20, 25] m/s
stage3_timer_seconds	60
stage3_pass_threshold	0.70
hint_available_stages	[1, 2]

Equation 2:Linear Motion Lab game template

Appendix D: Risk Analysis

Risk	Likelihood	Impact	Mitigation
NIE textbook content insufficient for SBERT fine-tuning (< 200 pairs)	Low	High	*Supplement with SciQ and augment NIE pairs through paraphrase generation*

SBERT fine-tuned model shows no improvement over BM25 baseline	Medium	Medium	Report BM25 as primary system; present SBERT as exploratory contribution with analysis of failure cases
flutter_unity_widget incompatibility on target Android version	Low	High	Test on minimum API 21 device early; maintain WebView-based fallback game if Unity integration fails
Google Colab session timeout during SBERT training	Medium	Low	Save model checkpoints after each epoch; use Colab Pro free trial if needed
Firebase Firestore free tier quota exceeded during testing	Low	Low	Cache Firestore reads locally in Flutter; batch writes to reduce quota usage
Insufficient student participants for usability pilot	Medium	Medium	Conduct evaluation with researcher, supervisor, and team members as proxy users; supplement with rubric assessment

Table 12:Risk Analysis